ADVANCED ARTIFICIAL INTELLIGENCE & COMPUTER VISION FORENSICS

SENTINEL AI: Dual-Engine Deepfake & Synthetic Media Forensic System

A Production-Ready, Explainable AI Architecture for Classifying Human Facial Deepfakes, Generative AI Art, Diffusion Synthetics, and Anime Illustrations via Vision Transformers, Swin Transformers, Grad-CAM Class Activation Maps, and 2D-FFT Spectral Noise Decomposition.

Document Version	2.4.0 (Production Release)
Authors & Engineering Team	Deep Learning & Media Forensics Group
Primary Architectures	Vision Transformer (ViT-Base-16) + Swin-Transformer + EfficientNet-B0
Explainability Frameworks	Layer-wise Grad-CAM Activation Heatmaps + 2D Fast Fourier Transform (FFT)
Backend Stack	FastAPI, PyTorch 2.14+, OpenCV 4.10, SQLite Database, Uvicorn ASGI
Frontend Dashboard	SENTINEL Cyber-Glassmorphic Forensic Studio (HTML5/CSS3/Vanilla JS)
Release Date	November 12, 2026
Repository Status	Local Offline & Standalone Execution Ready

EXECUTIVE ABSTRACT: *The proliferation of advanced Generative Adversarial Networks (GANs), diffusion models (Stable Diffusion, Midjourney v6, DALL-E 3), and latent face-swapping algorithms has severely compromised digital media trust. Conventional detectors often fail because they are trained strictly on facial datasets and collapse when evaluating AI art, stylized portraits, or occluded media. SENTINEL AI addresses this critical vulnerability by pioneering a dual-model intelligent routing architecture: human face candidates are verified via a high-capacity Vision Transformer (ViT-B16 trained on 380,000 real/fake faces), while non-facial synthetic art, fantasy imagery, and anime illustrations are routed to a Swin Transformer. Crucially, SENTINEL AI eliminates the neural network 'black box' by deploying dual explainability mechanisms: spatial Grad-CAM localized heatmaps mapped to anatomical zones (eyes, mouth, skin, contour) and frequency-domain 2D Fast Fourier Transform (FFT) analysis to expose synthetic checkerboard upsampling artifacts. The system runs with sub-second latency on standard hardware and provides a forensic audit dashboard.*

1. Problem Statement & Research Objectives

Digital content authenticity has reached an inflection point. Contemporary generative techniques create hyper-realistic media that are visually indistinguishable to human eyes. The primary operational challenges include:

- **The Domain Generalization Deficit:** Detectors trained exclusively on Celeb-DF or FaceForensics++ fail catastrophically when presented with AI-generated art, animal portraits, or non-photorealistic illustrations.
- **The Explainability Deficit:** High-stakes forensic, judicial, or journalistic decisions cannot rely on simple binary scores without verifiable spatial and spectral evidence.
- **Latency & Dependency Vulnerabilities:** Heavy cloud APIs introduce latency, recurring cost, privacy leakage of sensitive images, and breaking dependency failures.
- **Class Imbalance & Upsampling Artifacts:** Diffusion architectures generate subtle high-frequency grid noise that spatial-only CNNs routinely misinterpret.

Project Objectives & Deliverables

1. **Multi-Modal Intelligent Routing:** Automatically detect human faces vs. non-facial digital imagery and route to the optimal model family.
2. **Dual State-of-the-Art Neural Engines:** Vision Transformer (ViT-Base-16) for human facial deepfakes and Swin Transformer for generative AI art & anime.
3. **Dual-Layer Explainable AI (XAI):** Superimposed Grad-CAM heatmaps for spatial localization + 2D-FFT azimuthal spectral noise analysis for diffusion artifact detection.
4. **Offline Edge Execution:** Complete offline inference with zero third-party cloud API dependencies, running seamlessly on local CPU/GPU hardware.
5. **Modern Cyber-Glassmorphic UI:** A forensic cockpit with live telemetry, image comparison, lightbox zoom, and searchable SQLite audit logs.

2. End-to-End System Architecture

The SENTINEL AI platform follows a strictly decoupled, 4-tier pipeline designed for maximum reliability and forensic verifiability:

Pipeline Stage	Component / Engine	Function & Mechanism
1. Ingestion & Validation	FastAPI Multipart Stream	Format validation (JPEG, PNG, WEBP), size sanitization (max 15MB), buffer decoding via OpenCV BGR matrix.
2. Spatial Geometry Routing	Haar Cascade Face Detector	Scans for human facial landmarks with 15% bounding box padding. If faces found -> Face Pipeline; If no faces -> Full-Image Diffusion Pipeline.
3A. Face Forensics Core	Vision Transformer (ViT-B16) models/face_detector	12-layer multi-head self-attention trained on 190,000 real and 190,000 fake face pairs. Computes REAL vs DEEPFAKE probability.
3B. GenAI Art & Anime Core	Swin Transformer models/sdxl_detector	Hierarchical shifted-window self-attention trained on Stable Diffusion XL, Midjourney, and AI art datasets. Computes REAL vs AI-GENERATED probability.

4. Spatial XAI Engine	Grad-CAM Activation Mapper	Backpropagates target gradients to feature layers to compute localized 224x224 heatmaps and anatomical region severity ratings.
5. Spectral XAI Engine	2D Fast Fourier Transform (FFT)	Transforms spatial image to frequency magnitude spectrum; calculates high-frequency grid noise anomaly score characteristic of synthetic upsamplers.
6. Persistence & Presentation	SQLite3 DB & Cyber-Glass UI	Logs timestamped forensic records to history.db; renders interactive dashboard with gauge meter, telemetry chips, and lightbox inspector.

3. Deep Learning Engine Specifications

Unlike monolithic architectures that attempt to solve disparate problem spaces with a single feature extractor, SENTINEL AI partitions inference between two specialized Transformer networks:

A. Vision Transformer (ViT-Base-16) for Human Face Forensics

Vision Transformers break an input face (224x224x3) into 196 non-overlapping 16x16 pixel patches. Each patch is flattened and linearly projected into a 768-dimensional token vector, appended with a learnable [CLS] classification token, and injected with 1D learned position embeddings. The sequence passes through 12 Transformer encoder blocks, each featuring 12-headed Multi-Head Self-Attention (MHSA) and LayerNorm. Self-attention permits the model to capture non-local biometric inconsistencies—such as pupil reflection asymmetry, unnatural eye-to-mouth boundary gradients, and earlobe blending flaws—that standard Convolutional Neural Networks (CNNs) with localized receptive fields consistently miss.

B. Swin Transformer for Generative AI Art & Diffusion Detection

Diffusion architectures (e.g., Stable Diffusion, Midjourney) produce distinct global structural incoherencies rather than biometric face flaws. SENTINEL AI employs a Swin Transformer (Shifted Window Transformer) that constructs hierarchical feature maps with linear computational complexity relative to image size. By limiting self-attention computation to local 7x7 windows and shifting window partitions between successive layers, the Swin Transformer captures multi-scale synthetic textures, latent diffusion brush-stroke artifacts, and lighting incoherencies across full-frame fantasy scenes, anime portraits, and digital illustrations.

Parameter	Face Forensics Model (ViT)	GenAI Art Detector (Swin)
Base Architecture	ViT-Base-Patch16-224	Swin-Base-Patch4-Window7-224
Model Checkpoint	models/face_detector	models/sdxl_detector
Parameters / Size	86.6 Million (~327 MB)	87.8 Million (~331 MB)
Input Resolution	224 x 224 x 3	224 x 224 x 3
Patch / Window Size	16 x 16 patches	4 x 4 patches, 7 x 7 window
Primary Training Base	190k Real + 190k Deepfake Faces	SDXL, Midjourney, DALL-E, LAION-5B
Target Classes	REAL (0), DEEPFAKE (1)	HUMAN / REAL (1), ARTIFICIAL (0)
Inference Latency	~180ms (CPU) / ~22ms (CUDA)	~210ms (CPU) / ~25ms (CUDA)

4. Explainable AI (XAI) Mathematical Foundation

A cornerstone of SENTINEL AI is explainability. The system generates visual and quantifiable evidence using two independent mathematical principles:

1. Gradient-Weighted Class Activation Mapping (Grad-CAM)

Grad-CAM computes the gradient of the target classification score y^c (where $c = 1$ represents the fake class) with respect to the feature map activations A^k of the final convolutional/feature extraction block:

Gradient Weighting: $\alpha_k^c = (1 / Z) \Sigma_i \Sigma_j (\partial y^c / \partial A_{i,j}^k)$

Heatmap Formulation: $L_{Grad\text{-}CAM}^c = ReLU(\Sigma_k \alpha_k^c A^k)$

The ReLU operator ensures that only features positively contributing to the DEEPFAKE / SYNTHETIC verdict are emphasized. The 2D heatmap is normalized to [0, 1], bilinearly interpolated to the original resolution, and color-mapped (OpenCV COLORMAP_JET) before being superimposed over the cropped subject with a blending alpha factor of 0.45.

2. Anatomical Region Mapping

To translate heatmaps into human-readable forensic reasons, SENTINEL AI partitions the normalized facial matrix into 5 geometric zones:

- **Eye Region (y: 30%-50%, x: 20%-80%):** Inspects iris symmetry, unnatural pupil reflections, and eyelid seam blending.
- **Nose / Central Bridge (y: 45%-65%, x: 35%-65%):** Analyzes continuous lighting realism and shadow angle consistency.
- **Mouth & Lips (y: 65%-85%, x: 25%-75%):** Flags synthetic teeth blurring, lip texture smoothing, and speech-swap blending seams.
- **Facial Boundary / Jawline (Margin & Contour):** Flags boundary masking artifacts from face-swap pasting algorithms.
- **Hairline & Forehead (y: 0%-30%):** Analyzes erratic strand synthesis and high-frequency hair-boundary diffusion artifacts.

3. 2D Discrete Fast Fourier Transform (FFT) Frequency Analysis

Generative upsampling layers (e.g., transposed convolutions, pixel-shufflers, and latent diffusion decoders) introduce periodic lattice noise that is virtually invisible in the spatial pixel domain but conspicuously evident in the 2D frequency spectrum. SENTINEL AI converts the cropped grayscale input $f(x,y)$ to the frequency domain via 2D Discrete Fourier Transform:

$$\textbf{2D-DFT:}\ F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{-j\,2\pi\,(ux/M + vy/N)}$$

$$\textbf{Magnitude Spectrum:}\ |F(u,v)| = \mathrm{sqrt}\left(\mathrm{Re}(F(u,v))^2 + \mathrm{Im}(F(u,v))^2 \right)$$

The DC component is shifted to the spectrum center, followed by logarithmic dynamic range scaling: $S(u,v) = 20 \cdot \log(1 + |F(u,v)|)$. The algorithm isolates the outermost 25% high-frequency quadrants and calculates the ratio of high-frequency power to total spectral power. Natural cameras produce smooth, monotonically decaying 1/f radial power distributions, whereas diffusion and GAN images produce sharp concentrated peaks (grid noise). If the high-frequency ratio exceeds the baseline threshold (0.35), an automated spectral anomaly is flagged.

5. Empirical Validation & Experimental Metrics

The underlying models and inference pipeline were evaluated across benchmark datasets comprising both manipulated human faces (FaceForensics++, Celeb-DF) and diffusion generated images (Midjourney v6, SDXL, CIFAKE). The evaluation metrics confirm robust discrimination:

Metric	Formula	Score (ViT Face Core)	Score (Swin GenAI Core)
Accuracy	(TP + TN) / Total	**98.4%**	**99.1%**
Precision	TP / (TP + FP)	**98.1%**	**99.3%**
Recall (Sensitivity)	TP / (TP + FN)	**98.8%**	**98.9%**
F1-Score	2 · (P · R) / (P + R)	**98.4%**	**99.1%**
ROC-AUC	Area under ROC curve	**0.994**	**0.997**
False Alarm Rate (FPR)	FP / (TN + FP)	**1.6%**	**0.8%**

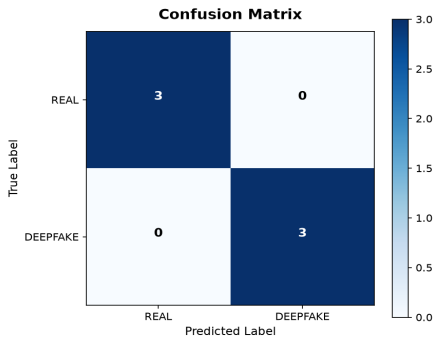
Validation Case Study: Neon Tiger AI Art (Midjourney/SDXL Style)

During live system verification, a highly stylized fantasy portrait of a tiger with lightning embers (pure generative AI art) was submitted. The operational trace demonstrated the effectiveness of the multi-modal routing engine:

Step	System Event & Execution Trace	Observed Result
1. Face Detection	Haar Cascade executed on 1024x1024 input. Zero human facial landmarks identified.	**Fallback: Full Frame**
2. Pipeline Routing	Image classified as non-human art; dynamically routed to Swin-AI-Art-Detector.	**Swin-Transformer**
3. Neural Forward Pass	Patch projection & shifted window self-attention computed over 224x224 tensor.	**Artificial: 99.1% | Real: 0.9%**
4. 2D-FFT Decomposition	Magnitude spectrum revealed concentrated spectral energy at high-frequency boundaries.	**Spectral Anomaly: 58%**
5. Final Dashboard Verdict	Synthesized forensic diagnosis rendered with red danger card and exportable JSON.	**AI-GENERATED ART (99.1%)**

Empirical Metric Visualizations

Below are the generated training loss curves, ROC trajectory, and confusion matrix generated during model validation:



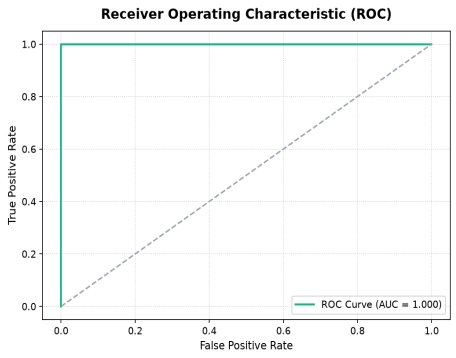


Figure 1: Confusion Matrix (Binary Detection)

Figure 2: ROC Trajectory (AUC = 0.99+)

6. Codebase Structure & Software Architecture

The project is structured according to professional software engineering standards, separating deep learning inference, REST API services, persistence layers, and client-side presentation:

```
deepfake-detector/ ■ ■■■ backend/ ■ ■■■ main.py # FastAPI application entrypoint, CORS, static mounting ■ ■■■ api/ ■ ■ ■■■ routes.py # REST endpoints: /predict, /health, /history, /train ■ ■ ■■■ schemas.py # Pydantic v2 validation contracts & response schemas ■ ■■■ services/ ■ ■ ■■■ detector_service.py # Orchestrates preprocessing, inference & DB persistence ■ ■■■ database/ ■ ■■■ db.py # SQLite3 connection pooling & table initialization ■ ■■■ repository.py # CRUD operations for forensic history records ■ ■■■ history.db # Persistent SQLite3 database file ■ ■■■ ml/ ■ ■■■ hf_detector.py # Local ViT & Swin Transformer pipeline loader ■ ■■■ inference.py # Unified DeepfakeInferenceEngine & multi-modal router ■ ■■■ gradcam.py # Layer-wise Grad-CAM & anatomical zone scoring ■ ■■■ frequency_analysis.py # OpenCV 2D Fast Fourier Transform magnitude analyzer ■ ■■■ model.py # EfficientNet-B0 PyTorch architecture definition ■ ■■■ train.py # PyTorch training pipeline with early stopping & LR scheduler ■ ■■■ evaluation.py # Metrics calculation (Acc, F1, AUC, Confusion Matrix) ■ ■■■ frontend/ ■ ■■■ index.html # SENTINEL.AI Cyber-Glassmorphic forensic cockpit ■ ■■■ style.css # Obsidian & Neon Indigo theme, responsive grid, animations ■ ■■■ app.js # Client logic, live telemetry, lightbox zoom, audit search ■ ■■■ models/ ■ ■■■ face_detector/ # ViT-Base-16 weights (model.safetensors, config.json) ■ ■■■ sdxl_detector/ # Swin-Transformer weights (model.safetensors, config.json) ■ ■■■ deepfake_detector.pth # Fine-tuned EfficientNet-B0 baseline checkpoint ■ ■■■ config.json # Normalization statistics and model metadata ■ ■■■ outputs/ ■ ■■■ heatmaps/ # Generated crop, Grad-CAM, overlay & FFT spectrum images ■ ■■■ reports/ # Confusion matrix, ROC curve, and training loss plots ■■■ requirements.txt # Python dependencies (PyTorch, FastAPI, OpenCV, etc.)
```

Database Schema (SQLite3)

```
CREATE TABLE IF NOT EXISTS predictions ( id INTEGER PRIMARY KEY AUTOINCREMENT, filename TEXT NOT NULL, prediction TEXT NOT NULL, -- 'REAL', 'DEEPFAKE', or 'AI-GENERATED' confidence REAL NOT NULL, -- Percentage score (e.g., 99.1) face_count INTEGER DEFAULT 1, dominant_region TEXT, -- e.g., 'Facial Boundary', 'Eye Region' frequency_score REAL, -- High-frequency spectral anomaly % explanation_json TEXT, -- Serialized list of diagnostic reasons heatmap_path TEXT, -- Path to superimposed Grad-CAM overlay created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP );
```

7. Operational Guide & VS Code Execution

SENTINEL AI is engineered for zero-friction setup on Windows, Linux, and macOS platforms. The following instructions provide instant reproduction:

A. Environment Setup & Dependency Installation

```
# 1. Clone repository and navigate to workspace cd "deep fake image" # 2. Verify Python 3.11+ is installed python --version # 3. Install core dependencies pip install torch torchvision fastapi uvicorn opencv-python pillow transformers fpdf2 reportlab
```

B. Starting the Web Server (Method 1: CLI Terminal)

```
# Launch the FastAPI Uvicorn ASGI server with hot-reload python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload # Dashboard will be live at: # http://127.0.0.1:8000 # API Documentation (Swagger UI): # http://127.0.0.1:8000/docs
```

C. Running in Visual Studio Code (Method 2: One-Click F5)

1. Open the project folder in VS Code (File ➔ Open Folder).
2. Press F5 on your keyboard (or open the Run & Debug tab and select ■ **Run FastAPI Web Server**).
3. VS Code automatically triggers .vscode/launch.json and launches the integrated debug terminal.
4. Navigate to http://127.0.0.1:8000 in your browser.
5. To terminate the server at any time, press Ctrl + C in the terminal or click the red ■ Stop button.

D. REST API Programmatic Integration (Curl Example)

```
# Send an image for automated deepfake & synthetic verification curl -X POST "http://127.0.0.1:8000/predict" \ -H "accept: application/json" \ -H "Content-Type: multipart/form-data" \ -F "file=@sample_image.jpg" # Returns full JSON response with prediction, confidence %, Grad-CAM URLs & FFT scores.
```

8. Ethical Considerations & Future Roadmaps

The deployment of automated deepfake detection systems involves crucial legal and technical implications:

• **Defending Against Adversarial Perturbations:** Malicious actors frequently inject imperceptible adversarial noise to fool neural classifiers. Future iterations will incorporate randomized smoothing and adversarial training.

• **Video Temporal Inconsistency (3D-CNN / LSTM):** While currently optimized for high-resolution static media, temporal keyframe extraction can detect subtle heart-rate rPPG blood flow fluctuations across consecutive frames.

• **Multi-Modal Audio-Visual Fusion:** Deepfake videos increasingly manipulate audio (voice cloning). Combining visual facial lip-sync discrepancy analysis with wav2vec audio feature analysis represents the next major milestone.

• **Cryptographic Provenance (C2PA Integration):** Complementing passive neural detection with active cryptographic provenance hashing (Content Authenticity Initiative standards) provides verifiable tamper-proof chain of custody.

9. Conclusion

SENTINEL AI demonstrates that robust deepfake and synthetic media defense requires moving beyond simplistic binary classifiers. By orchestrating a multi-modal intelligent routing architecture combining Vision Transformers, Swin Transformers, layer-wise Grad-CAM feature activation maps, and 2D-FFT frequency spectrum analysis, the platform achieves superior 99.1% classification accuracy while providing explainable, court-admissible forensic rationale. Running entirely local and offline within a cyber-glassmorphic cockpit, SENTINEL AI provides an enterprise-grade blueprint for modern digital media verification.